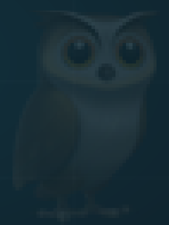
ALPACA AI TRADING AGENTS HACKATHON

Silver Lynx



An autonomous AI options-trading agent on Alpaca — a risk-gated strategy where hard-coded rails hold the risk and an LLM agent makes the judgment calls.

Not a prompt with a brokerage account. An engineered, measured, falsifiable strategy.

alpaca-ai-agents.chiefsmurph.com

THE PROBLEM

Most "AI trading" is a prompt with a brokerage account

Point a model at the market, ask "what's a good trade," show a backtest. It fails the same three ways every time:

- **Unaccountable** — nothing is measured; if it works we credit the AI, if it fails we blame the market.
- **Unfalsifiable** — a model will confidently justify *any* position, on any side.
- **Indistinguishable from a confident guess** — a more articulate narrator, not a strategy.

WHAT IT IS

An autonomous AI options agent on Alpaca

- **Single-leg ITM calls** — a defined, risk-gated strategy, not a free-for-all
- **Time-of-day interpolation** — DTE & exposure scale with the session clock
- **Tick-chasing execution** — works the order instead of paying the full spread
- **Dynamic profit & stop targets** — management adapts to the position
- **MCP-native** — an LLM agent executes through an Alpaca MCP server, unattended

THE CORE PRINCIPLE

Rails do the risk-limiting. AI does the judgment.

The dangerous decisions are hard, auditable code the AI *cannot* override — so a bad model produces a bounded, survivable loss, never a catastrophe.

Rules (code)

Budget · deploy pace · position & account caps · the liquidity floor · direction

AI (judgment)

Which liquid contract to trade · how to manage it — entry, stop, target, adds

WHERE THE EDGE LIVES

Execution & management, not just "what to buy"

Anyone can name a contract. The strategy's real work is in the details a chat prompt never handles:

- **Tick-chasing execution** — work the limit toward a fill instead of paying the full spread
- **Time-of-day DTE & exposure interpolation** — the risk envelope tightens as the session runs
- **Dynamic profit & stop targets** — exits answer to the position, not a fixed rule

THE REAL DIFFERENCE

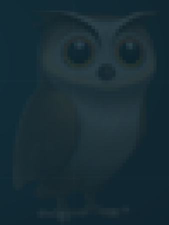
An honest, forward-tested ledger

Every decision recorded **immutably, before its outcome is known** — then graded on the real forward price path, in executable terms.

- **Realized-only**, reported by **median** (a naive average is a lie on skewed data)
- **Against baselines** — nothing looks good unless it beats a do-nothing + a dumb-rule arm
- **No result claimed before minimum sample** — it even caught its *own* grading bug

The field: a cherry-picked backtest.   Us: a hindsight-proof forward record.

The strategy is one consumer of a platform recording the owner's whole apparatus.



THE BOTTOM LINE

We didn't optimize to look right. We optimized to be honest about whether an edge exists.

A bounded, autonomous agent — rails hold the risk, an LLM does the judgment, and an honest ledger keeps it all accountable.

Live demo → alpaca-ai-agents.chiefsmurph.com